

ATLEE AGCM, AB, Canada

WMO: 719860

Lat: 50.810N

Lon: 111.006W

Elev: 795

StdP: 92.13

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.2	-26.5	-32.4	0.2	-29.1	-29.3	0.3	-26.4	14.9	-1.8	13.2	-0.9	3.0	170	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	31.2	16.3	29.2	16.0	27.2	15.7	18.6	25.9	17.7	25.3	16.9	24.6	5.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.7	21.3	15.1	11.8	20.2	14.0	11.0	19.5	56.0	25.7	53.2	25.2	50.3	24.6	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.5	10.0	-33.5	34.6	2.4	1.8	-35.3	35.9	-36.7	37.0	-38.1	38.0	-39.8	39.3		
(5)			-33.6	20.8	2.4	1.3	-35.4	21.7	-36.8	22.5	-38.2	23.3	-39.9	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-9.1	-10.5	-2.9	4.9	11.5	15.5	18.7	18.0	13.0	5.1	-3.3	-10.2
	DBStd	12.27	8.08	8.27	8.05	4.92	4.32	3.41	3.11	3.43	5.09	5.28	7.59	7.80
	HDD10.0	2981	591	576	399	163	33	3	0	0	26	164	400	625
	HDD18.3	5226	849	809	657	403	214	98	33	49	171	409	648	883
	CDD10.0	908	0	0	1	10	80	166	271	248	117	14	2	0
	CDD18.3	112	0	0	0	0	3	12	46	39	12	0	0	0
	CDH23.3	1664	0	0	0	4	88	181	633	535	219	3	0	0
	CDH26.7	530	0	0	0	0	16	43	216	179	76	0	0	0
Wind	WSAvg	4.9	4.8	4.5	5.1	5.6	5.2	5.0	4.6	4.4	4.7	5.0	4.8	4.5
Precipitation	PrecAvg	354	12	8	15	28	46	81	42	37	32	17	15	12
	PrecMax	528	33	22	41	81	103	202	121	101	83	44	42	32
	PrecMin	151	3	0	1	1	7	27	6	2	4	2	1	2
	PrecStd	95	9	7	10	19	30	42	34	23	23	11	11	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	7.9	16.6	23.1	28.9	30.9	33.9	33.2	32.1	23.1	17.4	7.4
		MCWB	3.8	3.5	8.0	10.8	14.2	16.1	17.3	16.1	15.3	12.9	8.8	3.9
	2%	DB	5.4	4.9	13.5	19.8	25.8	27.6	31.3	30.9	29.1	19.2	12.5	4.5
		MCWB	2.5	1.8	6.5	9.3	13.0	15.5	17.1	16.2	14.6	10.8	6.4	2.1
	5%	DB	3.7	2.6	10.7	16.8	23.4	25.3	29.1	28.6	25.6	16.5	9.1	2.9
		MCWB	1.6	0.7	5.2	7.7	11.8	15.1	16.8	16.0	13.7	9.2	4.6	0.8
	10%	DB	2.0	0.6	7.1	14.0	20.7	23.2	26.9	26.4	22.7	13.6	5.9	0.7
		MCWB	0.5	-0.7	3.3	6.2	10.9	14.0	16.4	15.7	12.7	7.7	2.7	-1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.4	8.6	11.2	14.9	18.8	20.5	19.3	16.9	13.4	9.2	4.4
		MCDB	8.0	7.6	15.4	21.3	26.7	25.5	28.6	26.1	26.1	21.4	17.1	6.6
	2%	WB	2.7	2.0	7.2	9.7	13.7	17.3	18.8	18.3	15.5	11.5	6.9	2.2
		MCDB	5.1	4.3	12.8	18.9	23.9	24.4	26.3	26.0	25.4	18.1	11.6	4.4
	5%	WB	1.7	0.9	5.2	8.1	12.7	16.1	18.1	17.3	14.4	9.6	4.7	0.9
		MCDB	3.6	2.6	9.9	15.3	21.5	23.0	25.2	25.3	24.1	15.4	8.3	3.1
	10%	WB	0.4	-0.8	3.3	6.7	11.5	14.9	17.3	16.4	13.2	8.1	3.1	-1.0
		MCDB	2.0	0.5	6.9	13.0	19.0	21.5	24.8	24.3	22.2	13.1	5.6	0.6
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.8	10.5	12.3	13.3	12.2	13.5	13.4	13.4	12.0	10.0	8.8
		MCDBR	9.5	11.2	14.9	17.5	17.3	15.2	17.1	17.4	18.5	16.1	13.7	10.6
		MCWBR	7.4	8.7	8.6	8.7	7.4	6.4	6.1	6.4	7.4	8.6	8.4	8.5
	5% WB	MCDBR	9.6	10.8	14.2	16.3	15.2	13.7	14.1	14.5	17.1	15.2	13.2	10.7
		MCWBR	7.7	8.6	8.6	8.5	6.8	6.7	6.2	6.0	7.3	8.2	8.6	8.6
Clear-Sky Solar Irradiance	taub		0.266	0.282	0.293	0.349	0.375	0.367	0.364	0.385	0.333	0.292	0.263	0.256
	taud		2.375	2.401	2.457	2.302	2.273	2.356	2.388	2.296	2.428	2.519	2.485	2.382
	Ebn at Noon		778	862	919	891	879	888	885	844	859	844	787	735
	Edh at Noon		61	78	89	117	126	118	112	117	92	69	54	52
All-Sky Solar Radiation	RadAvg		1.18	2.19	3.57	4.91	5.86	6.20	6.71	5.44	3.91	2.39	1.30	0.91
	RadStd		0.12	0.20	0.25	0.36	0.39	0.37	0.31	0.36	0.34	0.27	0.11	0.08

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page